Campus Item Exchange

Phases 4 and 5 Architecture and Detailed Design | 30 October 2026 | Academic prototype

Purpose and scope

An internal campus system for finding, requesting, transferring, maintaining and tracking reusable assets. The prototype covers one campus, two sample departments and about 20 labelled items. AI supports people with advice; permissions, stock, approvals and calculations remain controlled by application rules.

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Review and individual contributions

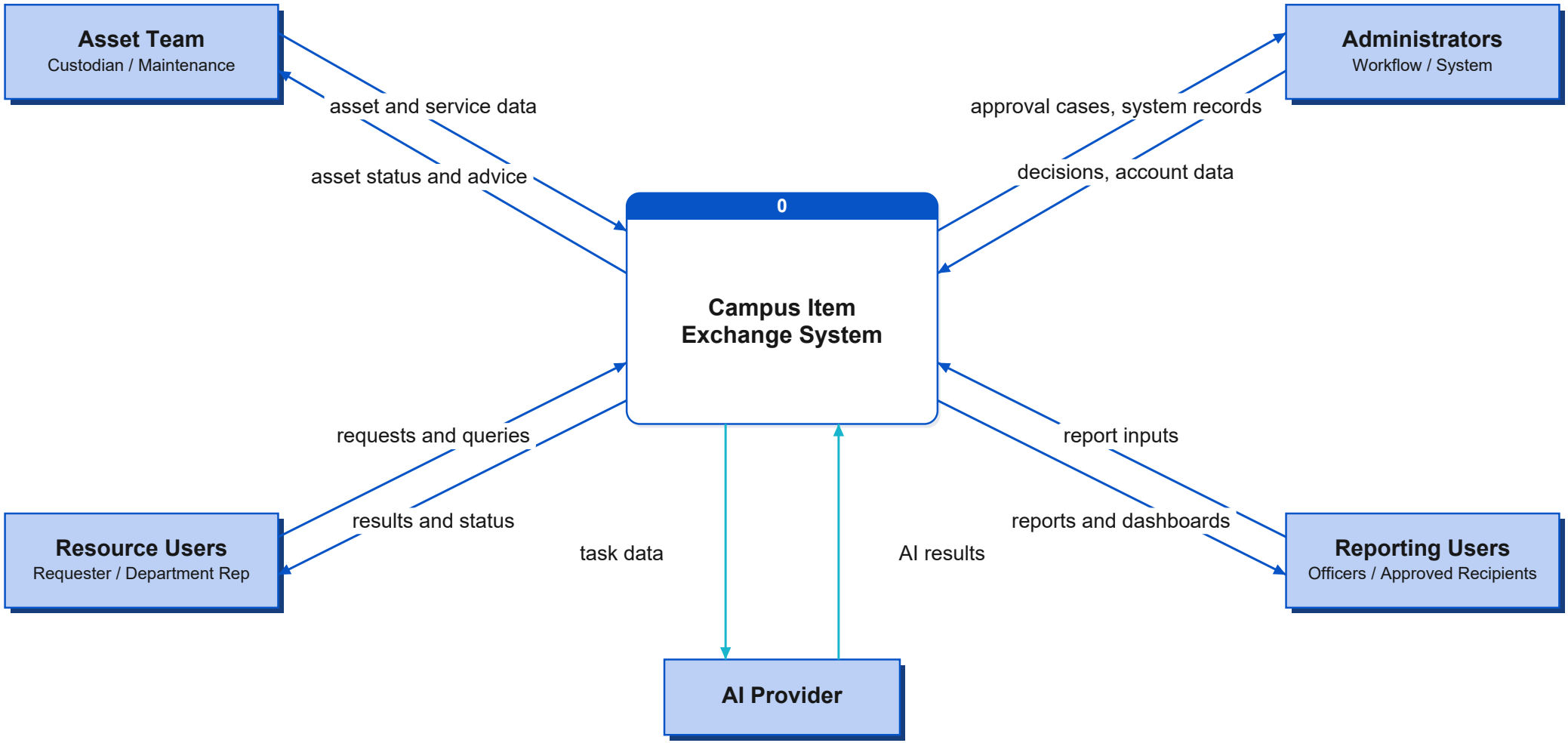
The four members and IDs match Phases 1-3. Each student must review and revise their contribution and commit it using their own identity. This document does not assert that individual GitHub commits or implementation tests have been completed.

Requirements and document status

Phase 3 requirement IDs appear in every section. Example records, AI outputs and screen values are synthetic. Deadlines, review gates and acceptance checks are proposed design settings; measured implementation results belong in later test records. This covers Phases 4 and 5.

Baseline: Campus Item Exchange Phase Three Requirements Document, 4 October 2026; Phase 4 Architecture Design lecture, Fall 2026.

Campus Item Exchange context diagram

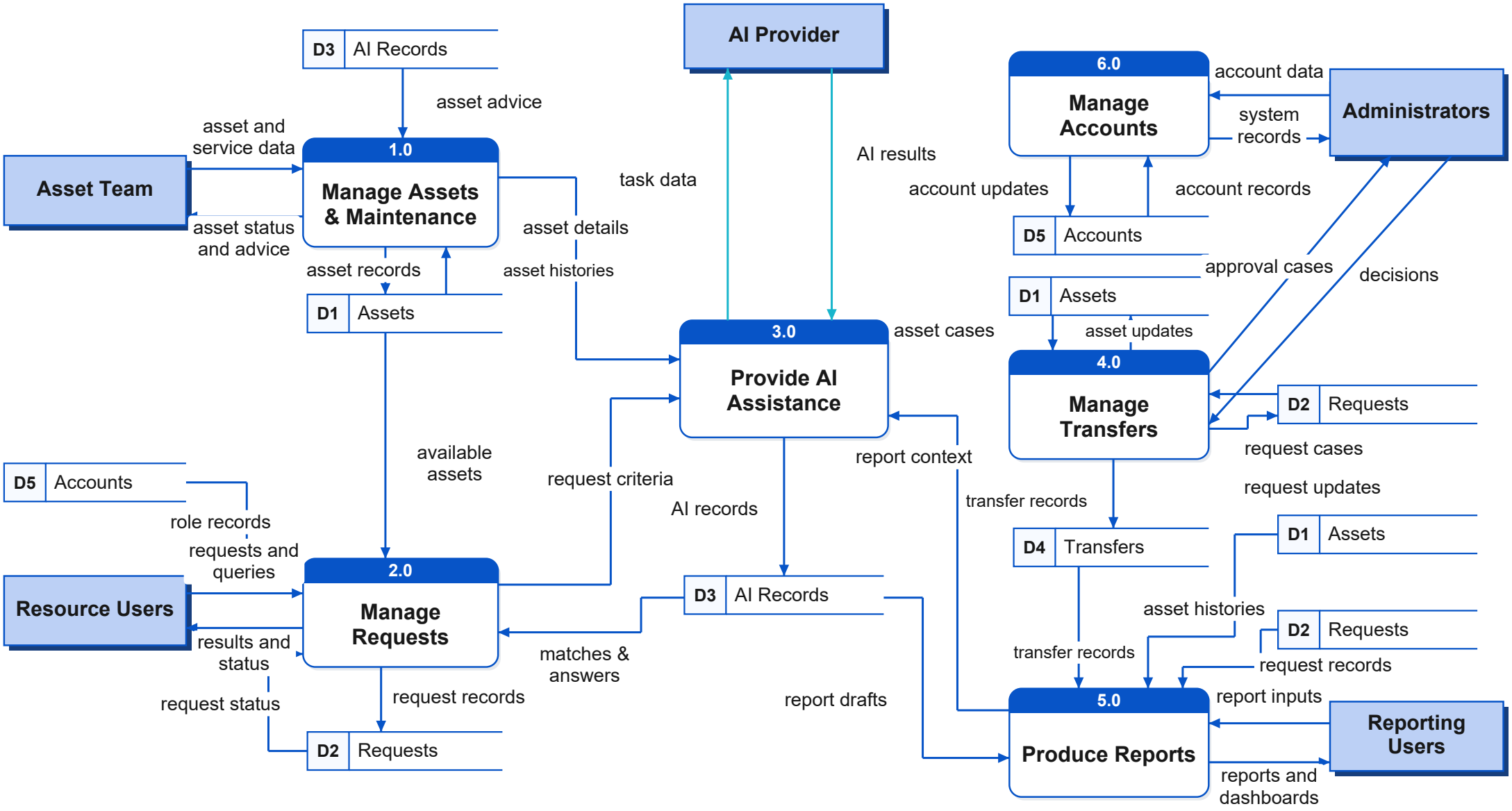


One process (0), grouped noun flows, no internal steps or data stores.

Reporting officers are Procurement, Finance and Sustainability. Grouping roles does not combine their permissions.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

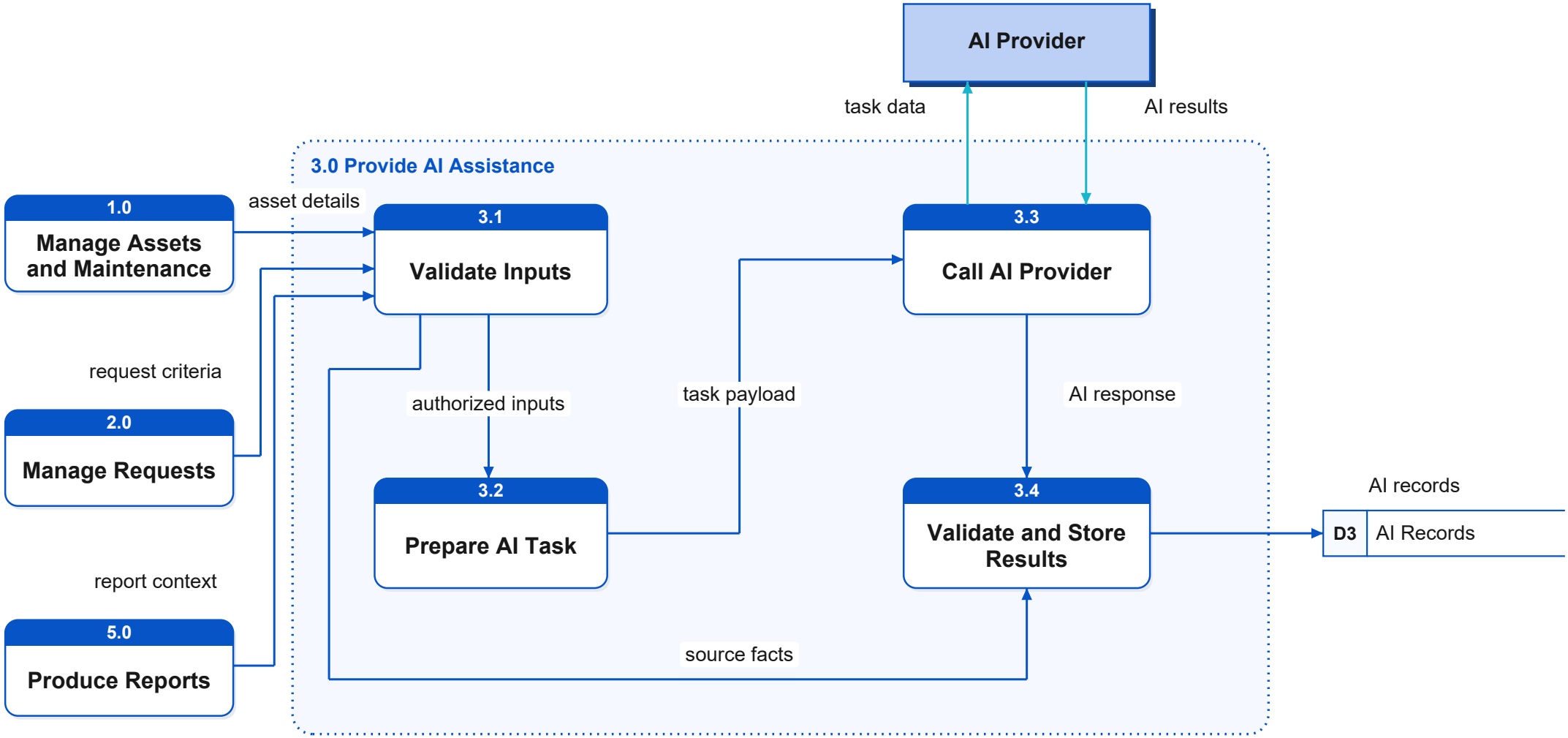
Campus Item Exchange Level 1 DFD



Repeated D-numbers refer to the same store. Maintenance is included in 1.0; approvals and disposition are included in 4.0.
Role families cover the full project. Sign-in, audit and notifications remain shared controls; permissions stay separate.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 2 of 3.0



The six flows crossing the dotted boundary are exactly the six flows of 3.0 on Level 1.

3.0 is expanded because it handles untrusted AI responses. Validated results or failure status are stored before returning to callers.

Phase 3 drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR13; BR01-BR07.

Campus Item Exchange Level 1 balancing table

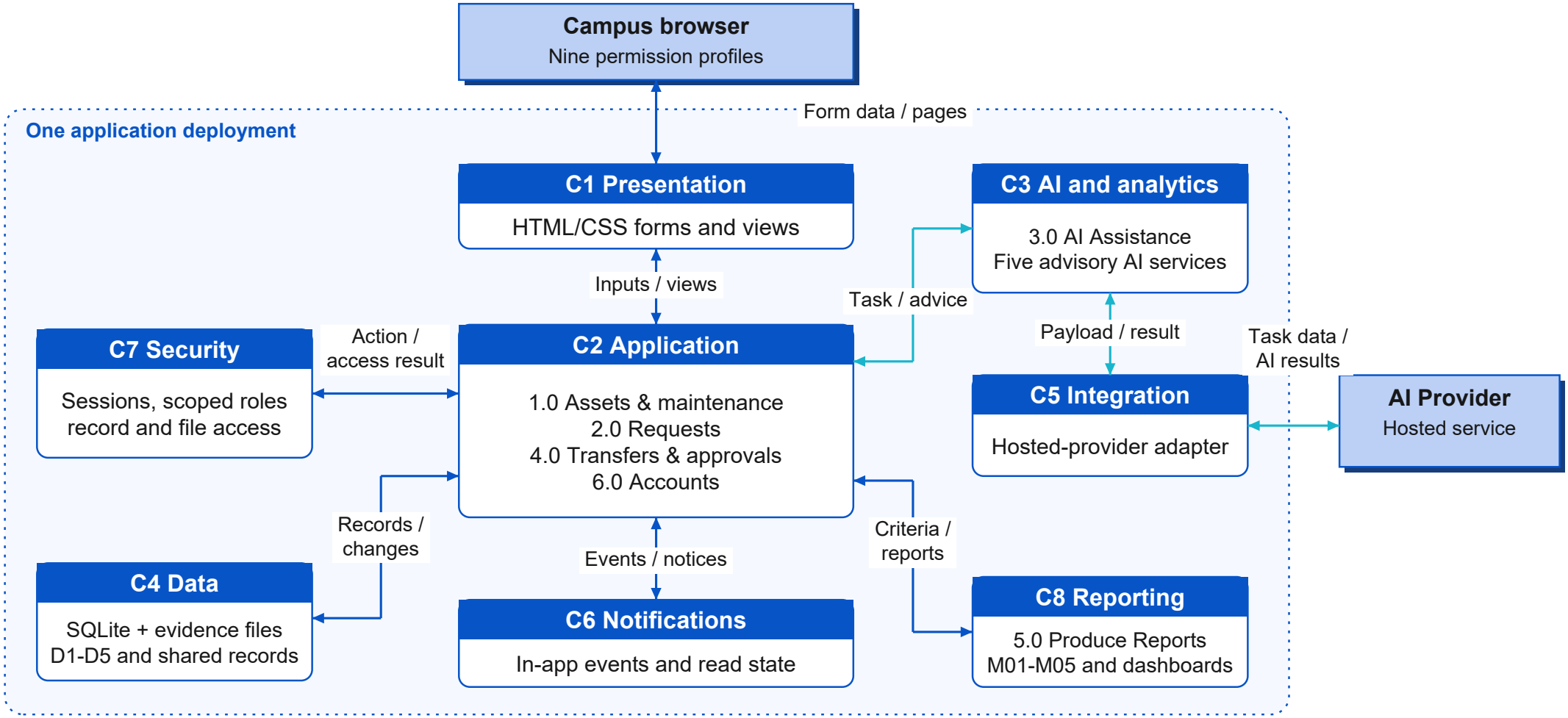
#	Context flow	External entity	In/out	Level 1 process	OK
1	asset and service data	Asset Team	in	1.0	✓
2	asset status and advice	Asset Team	out	1.0	✓
3	requests and queries	Resource Users	in	2.0	✓
4	results and status	Resource Users	out	2.0	✓
5	decisions, account data	Administrators	in	4.0, 6.0 (split)	✓
6	approval cases, system records	Administrators	out	4.0, 6.0 (split)	✓
7	report inputs	Reporting Users	in	5.0	✓
8	reports and dashboards	Reporting Users	out	5.0	✓
9	task data	AI Provider	out	3.0	✓
10	AI results	AI Provider	in	3.0	✓

VERIFY 10 of 10 match: same name, same direction, same entity. Grouped flows (5, 6) split at Level 1.

In/out is relative to the whole system. Flows 5 and 6 split between 4.0 Manage Transfers and 6.0 Manage Accounts.

3.1 Campus Item Exchange architecture view

Modular monolith: one Flask application, a private data environment and one external AI provider.



Routes call services; services use scoped repositories and coordinate AI and reports. Security applies to every server component.

Manual search, requests, asset history and numeric reports remain available during an AI failure. Paired labels show exchanged data.

Drivers: FR01-FR18; AI01-AI08; NFR03, NFR05-NFR06, NFR09-NFR11; BR01-BR07.

3.2 Architecture decision records

Each record compares two options, explains the choice and trade-off, and states a planned verification check.

ADR-01 Architecture style

Drivers: FR01-FR18; NFR05, NFR12; BR01-BR07.

Option A
Layered monolith: one application organized by presentation, business logic and persistence.

Option B
Modular monolith: one application organized around business responsibilities, with shared services.

Choice We choose a modular monolith because six DFD responsibilities give the team clear module boundaries while reservations and transfers can share local transactions.

Give up We give up the simpler layer-only organization. Module boundaries need discipline; deployment, restarts and scaling remain shared.

Check Trace 1.0-6.0 to their owning components. Check module interfaces and test reservation and transfer rollback without partial changes.

ADR-02 Presentation and application stack

Drivers: Phase 3 Section 1 baseline; FR11, FR16; NFR02-NFR04, NFR07-NFR08, NFR12.

Option A
Flask with server-rendered HTML/CSS, ordinary forms and limited browser scripting where needed.

Option B
Flask JSON API plus a separate React browser application, with coordinated client and API behavior.

Choice We choose Flask with HTML/CSS because it preserves the Phase 3 baseline and keeps the four-person prototype manageable. All forms call the same authorized application services.

Give up We give up a separately maintained client application and some browser interaction flexibility; additional clients would need API endpoints.

Check Test login, request and approval flows in the recorded Chrome and Firefox versions at 1280 x 720, including keyboard use and forbidden direct requests.

3.3 Data and AI decision records

The technology choices preserve the Phase 3 prototype baseline. Provider capabilities still require confirmation.

ADR-03 Prototype data and evidence storage

Drivers: Phase 3 Section 1 baseline; FR05, FR08, FR12-FR13; NFR05, NFR09, NFR15.

Option A
SQLite and protected evidence files, with scoped repositories, explicit transactions and coordinated snapshots.

Option B
PostgreSQL and protected evidence storage, with a separate database service to configure and recover.

Choice We choose SQLite because the baseline is one campus, two sample departments and about 20 items. Transactions must recheck availability and prevent duplicate reservations or handovers.

Give up We give up an independently deployed database service. The prototype must pass the required concurrent-write and growth checks before acceptance.

Check Race requests for the last unit; retry a handover; restore database and evidence within 10 minutes. Run NFR15 with 200 items and 10 sessions.

ADR-04 AI gateway and provider integration

Drivers: AI01-AI08; NFR03-NFR04, NFR06, NFR10, NFR14; FR05, FR17.

Option A
One internal AI gateway and provider adapter: classification, semantic matching, ranking, action advice and LLM assistance/report drafts.

Option B
Direct provider calls in each feature module, with separate validation, privacy, timeout and fallback handling.

Choice We choose one gateway because shared task contracts keep authorized payloads, output validation and provenance consistent. AI suggests; application rules and human approvals control changes.

Give up We give up direct feature-specific provider access and accept a common dependency for all AI features. Contract changes require coordination.

Check Test all five services with success, timeout and invalid output. Enforce 15 seconds per AI call; preserve manual workflows and model/review records.

AI-use statement: AI assisted with drafting, figures and consistency checks. Team review and implementation testing are pending.

4.1 AI service plan

A = ask and wait. All services run through 3.0 Provide AI Assistance and the shared provider adapter.

Service	Pattern	Why	Time limit	If down or invalid	If unsure
Asset classification AI02	A	Custodian is editing one asset and can continue manually.	8 s	Preserve draft; select category and tags manually. Explain unsupported evidence.	Below 0.80 or confidence Unavailable: manual-review warning. Never pre-accept; unknown facts stay Unknown.
Semantic matching AI01	A	Small dataset; compare current authorized candidates without a maintained vector index.	At most 6 s; within shared 12 s	Keep need and filters; show labelled keyword/rule results. Skip AI ranking after matching failure.	After hard checks, compatibility below 75/100 means weak match. Unavailable confidence needs review; no default choice.
Ranking and explanations AI06	A	User needs one ordered list from the same eligible candidate set.	At most min(6 s, shared time left)	Keep candidates in labelled rule order; show known factors without invented AI scores.	No top-pick emphasis if top score <75, top-two gap <10, required fact missing or confidence <0.80/Unavailable.
Sustainable-action advice AI03	A	User asks about one asset; its ordinary service/disposition form remains available.	12 s	Preserve evidence and use normal maintenance/disposition forms; old advice shows its date and stale warning.	Below 0.80, Unavailable confidence or missing evidence: qualify advice. No invented saving or safety clearance; human approval stays required.
LLM assistant and report drafts AI04-AI05	A	Person requests an answer or optional draft; numeric reports appear independently first.	12 s per explicit answer or separate draft	Retain question; offer normal screens. Keep numeric report visible and label the narrative unavailable.	Cite sources or say unavailable. Reject false IDs, policy or totals. <0.80/Unavailable requires review; requests remain previews until confirmed.

Budgets include retrieval, provider transport and validation. The Phase 3 limit is 15 seconds per AI call.

Drivers: AI01-AI08; FR01-FR05, FR09-FR10, FR13, FR18; NFR03, NFR06, NFR10, NFR13-NFR14; UC09-UC13.

4.2 Shared AI controls and verification

Thresholds are proposed prototype review gates. Confidence and compatibility have different meanings.

Deadline

Every budget includes scoped retrieval, provider transport and validation. Matching and ranking share one 12-second deadline: matching <=6 seconds, ranking <=min(6, remaining). No fresh deadline starts between stages. NFR06 remains 15 seconds per AI call.

Failure and capacity

No automatic retry or unbounded waiting. Return AI busy promptly if no configured capacity is available. Reserve web capacity for ordinary screens. A new attempt needs an explicit action; late results cannot replace a timeout or change records.

Confidence

Use a numeric confidence only if its origin and 0-1 scale are documented for that task. The proposed warning threshold is 0.80. Otherwise show Unavailable and require manual review; never invent confidence or treat an LLM's self-reported number as calibrated accuracy.

Compatibility

The proposed normal-match threshold is 75/100. Top-pick emphasis also needs a 10-point gap when two candidates exist, complete required compatibility facts and no unresolved confidence warning. These are display/review gates, not accuracy claims or permissions.

Validation and authority

Check current identity, record scope, source revision, response schema, IDs, categories, facts and report totals. Recheck stock and safety when a user confirms. No score can override a failed rule, authorize a disposition or clear an unsafe asset.

Provenance

Store input references/snapshots, time, provider, model/version or Version unavailable, prompt version, output, confidence or Unavailable, validation outcome, and accepted/edited/rejected/overridden decision with reason. Exclude passwords, session secrets and AI keys.

Planned acceptance checks

Demonstrate successful results for all five services, not only fallback screens. Test at least 8/10 suitable top-three matches and 8/10 correct labelled classifications, including supported image/PDF input.
Check ten explanations against real factors; test department-name swaps, fabricated IDs, wrong totals, revoked roles, stale inputs and late/duplicate responses. Record tester, environment, date, expected/actual outcome and evidence.
Before acceptance, confirm provider/model versions, evidence handling, documented confidence semantics and actual capacity. Numeric reports remain available before a separately requested narrative draft.

AI07-AI08; FR12, FR16-FR17; NFR03-NFR06, NFR10, NFR13-NFR14; BR01, BR04-BR07

4.3 AI pattern decision records

A = ask and wait; B = compute ahead; C = queue and notify. Each service has a real alternative and a planned check.

Record / service	Two options	Choice and because	What we give up	Planned check
ADR-05 Classification	A: interactive call; B: recompute after each asset revision.	We choose A because the custodian reviews fields while editing.	We give up ready-made suggestions and completion after leaving the editor.	Test valid text/image/PDF, unsupported evidence, timeout and retained draft.
ADR-06 Semantic matching	A: compare on demand; B: maintain versioned representations on save.	We choose A because about 20 items do not yet justify index and invalidation handling.	We give up instant reuse of precomputed vectors; revisit if growth/latency checks fail.	Use ten known needs; test permission changes, stale stock and matching timeout.
ADR-07 Ranking	A: rank the current list; C: queue ranking and notify later.	We choose A because the requester is waiting to choose from one current eligible list.	We give up deferred results after the user leaves and accept a bounded wait.	Test top-two ties, unknown factors, department swaps and the shared deadline.
ADR-08 Sustainable-action advice	A: explicit foreground advice; C: queue and notify when ready.	We choose A because one-asset reviews fit the prototype and manual forms provide continuity.	We give up advice that completes after navigation away and accept a short wait.	Test missing evidence, timeout, stale advice and unchanged approval/safety controls.
ADR-09 Assistant and report drafts	A: separate requested answer/draft; C: queue longer generation and notify.	We choose A because it keeps one interaction path while numeric reports remain immediately usable.	We give up deferred completion and hidden long-running generation.	Test cited answers, exact request preview, mismatched totals and draft failure after report display.

Drivers: AI01-AI08; NFR01, NFR03, NFR05-NFR06, NFR10-NFR11, NFR14-NFR15; UC09-UC13. All settings require validation.

5.1 Human review and approval points

AI may suggest, rank, explain or draft. A permitted person decides; server checks remain mandatory.

Point	Who decides	What they review and confirm	Why / required record
H1 Metadata	Assigned asset custodian	Review entered fields, evidence, suggested category/tags, confidence and missing facts. Accept, edit or reject; confirm final fields before saving.	Risk: invented or incorrect metadata. Record input revision, reviewed values, AI provenance and edit/override reason. FR01-FR02; AI02, AI07-AI08.
H2 Request or chat preview	Requester / department representative, within scope	Review exact asset, quantity, destination, date, need and stock. Explicitly confirm: normal rules create an atomic hold. Choosing a match alone reserves nothing; unmatched needs may stay Seeking.	Risk: stale or unauthorized allocation. Record exact preview, creator, stock check, resulting request/hold or failure and AI review. FR03-FR05; AI01, AI04, AI06-AI08; BR01-BR02.
H3 Publication	Workflow administrator; no self-approval	Review release proposal, owner, condition, serviceability and evidence. Approve or reject with a reason; only approved, serviceable active units may be published.	Risk: publishing unsafe or unreleased stock. Record actor, time, reason and old/new publication state. FR02, FR17; BR01, BR04; NFR13.
H4 Reservation / transfer decision	Workflow administrator; no self-approval	Review purpose, destination, quantity, hold/expiry and current eligibility. Approval retains the same hold and changes no ownership. Rejection releases stock once.	Risk: unsupported authority or double deduction. Record case/hold IDs, actor, time, reason and status transition. FR05, FR07, FR17; BR02-BR04; NFR05.

Every gate records actor, time, case ID, reason and the previous/new state. No one approves their own request.

Drivers: FR01-FR08, FR12-FR13, FR16-FR17; AI01-AI02, AI04, AI06-AI08; BR01-BR04; NFR03, NFR05, NFR13.

5.2 Physical actions, reporting and accounts

Human judgment and evidence control changes that AI cannot authorize or verify.

Point	Who decides	What they review and confirm	Why / required record
H5 Collection and receipt	Source and receiving custodians	Source records collection arrangements and evidence. Receiver confirms the full approved receipt. Only then does one transaction update ownership/custody. Refused or overdue receipt enters dispute; units stay blocked.	Risk: unproven physical handover. Record both actors, times, evidence, quantity/condition and old/new custody, owner, location and hold. FR08, FR13, FR17; BR02-BR06.
H6 Repair / safety	Custodian authorizes repair; assigned maintenance staff records inspection	Review defect, job state, costs or Unknown and AI advice. Follow service states and record human inspection clearance before restoring serviceability. AI cannot clear an asset.	Risk: unsafe units returning to use. Record authorization, inspection evidence, service actions, condition and advice override reason. FR09, FR14; AI03; NFR13; UC07, UC11.
H7 Disposition	Custodian proposes/executes; workflow administrator approves	Review unallocated units, holds, condition, costs and evidence. Approve/reject with a reason. Retirement blocks use; final disposition requires execution evidence. No self-approval.	Risk: irreversible loss or disposal of held stock. Record proposal, review, approval, actors/times, execution evidence and lifecycle transition. FR10, FR13, FR17; BR04-BR07.
H8 Reporting	Procurement, finance or sustainability officer; workflow admin for operations	Validate only assigned inputs. Review period, record links, calculated totals, coverage and optional AI draft. Reject mismatched figures; numeric reports remain usable without AI.	Risk: invented impact or wrong period. Record input changes, contributing IDs, calculation basis and narrative review. No additional report-approval state is introduced. FR12, FR18; AI05; BR07; NFR11.
H9 Accounts / roles	System administrator	Review identity, active status, profile, department and custody. Confirm changes; reassign custody before deactivation. This profile alone cannot approve business cases.	Risk: excess access or orphaned custody. Record target account, actor, time, reason and old/new role, department and active state. Exclude secrets. FR11-FR12, FR14, FR16; BR06.

Workflow administrators resolve receipt disputes: acknowledged receipt completes; evidenced physical return permits cancellation and stock restoration.

Drivers: FR08-FR18; AI03, AI05, AI07-AI08; BR02-BR07; NFR03-NFR05, NFR11, NFR13; UC06-UC08, UC11, UC13.

5.3 Human review design decision

The interaction changes with the evidence and authority needed at each gate.

Record ADR-10	Stage-specific human review
Drivers	FR07-FR08, FR10, FR17; AI07-AI08; BR04-BR06; NFR13.
Option A	One generic human decision form for all approval gates, with case-specific validation.
Option B	Stage-specific review forms for metadata, requests, approvals, receipt and disposition, using shared authorization and audit services.
Choice and because	We choose stage-specific review forms because each gate needs different evidence, role checks and consequences. Explicit confirmation invokes the ordinary transaction; AI never applies an action.
What we give up	We give up one reusable generic screen and accept more form states and validation paths.
Planned check	Try a self-approval, an AI-only submission, a receipt without evidence, an unsafe unit and a duplicate confirmation; verify the required gates and atomic effects.

Failure and concurrency behavior

If permission, state, safety, stock or evidence changes before confirmation, reject the action and preserve valid input. Failed writes leave no partial update; repeated confirmation returns the existing result. A record correction appends a linked entry rather than erasing history.

Drivers: FR05-FR08, FR10, FR12-FR13, FR16-FR17; AI07-AI08; BR04-BR06; NFR05, NFR13.